

Grade 7
Unit 9
Lesson 8 Practice

Name Answer Key
Date _____
Period _____

Write an equivalent expression using the associative, commutative, or distributive property. State the property used. **answers may vary**

1. $6x - \frac{2}{7}(21x - 35)$

$6x - 6 + 10$

distributive property

2. $(11x + 15) - 3x + 7$

$(15 + 11x) - 3x + 7$

commutative property

3. $\left(-\frac{3}{5}y + 2\right) \cdot 20$

$20 \cdot \left(-\frac{3}{5}y + 2\right)$

commutative property

4. $-24\left(\frac{3}{4}x + 1\right) + (18 - 3x)$

$-18x - 24 + (18 - 3x)$

distributive property

5. If expressions are not equivalent, then they will

- ☐ always
- ☐ sometimes
- ☒ never

have the same terms when they are simplified using properties of operations.

Use properties of operations to determine if the expressions are equivalent.

6. $0.5x - 0.1(5x - 2)$ and $7x - (-2 + 7x)$

$$0.5x - 0.5x + 0.2 = 0.2$$

$$7x + 2 - 7x = 2$$

not equivalent

7. $(4x + 12) - 6$ and $7x - 3(x - 2)$

$$4x + 12 - 6 = 4x + 6$$

$$7x - 3x + 6 = 4x + 6$$

equivalent

8. $2\frac{1}{5}x - 8 + x$ and $\frac{16}{5}\left(x - \frac{5}{2}\right)$

$$2\frac{1}{5}x + x - 8 = \frac{16}{5}x - 8$$

$$\frac{16}{5}x - \frac{16}{2} = \frac{16}{5}x - 8$$

equivalent

9. $0.3 + 0.1(17x - 4)$ and $2x - \left(\frac{3}{5}x - \frac{2}{5}\right)$

$$0.3x + 1.7x - 0.4 = 1.7x - 0.1$$

$$2x - \frac{3}{5}x + \frac{2}{5} = 1.4x + 0.4$$

not equivalent

10. Expressions that are equal are not equivalent.

- ☐ True
☒ False